Registration Number: 23BAI11298
Email: aldrinjoan6@gmail.com
aldrin.23bai11298@vitbhopal.ac.in
Phone: +91 8220863200
https://github.com/Aldrin-Joan
http://www.linkedin.com/in/aldrin-joan-pandian-w-08215028a

Aldrin Joan Pandian W

Programming Languages: Java, C++, Python
Mobile Development: Android Development, Flutter
Machine Learning & AI: Natural Language Processing (TF-IDF, LDA, NMF, Semantic Search), Machine Learning, Generative AI (GPT-5, LLM fine-tuning), Multi-Agent Systems, Prompt Engineering
Web Development: Web Development (React JS), API Integration, JavaScript
Cloud & DevOps: Azure Deployment, Cloud-based AI applications
Tools & Frameworks: LangChain, LangGraph, LangSmith, YOLO model, Responsive Design, Real-time Data Handling, File Processing, Automation
Certification:

- EC Council Certified Ethical Hacking V12 – EC Council (2023-27)
- AI Engineering – IBM

EDUCATION

Board	Tenure	Educational Institution	CGPA/Percentage
B. Tech (CSE) AI&ML	September 4, 2023 - Ongoing	VIT Bhopal University, Bhopal	8.63/10
Class XII	May 10, 2023	Carmel Garden Matric Hr Sec School, Coimbatore	75.0%
Class X	May 28, 2021	Carmel Garden Matric Hr Sec School, Coimbatore	100%

ACADEMIC PROJECTS

End-to-end Developer

- **GraviGauge – VS Code Extension (Antigravity Quota Monitor) (2026)**
  - **Description:** Designed and developed a production-ready VS Code extension that monitors real-time AI model quota usage from the local Antigravity application. Reverse-engineered Antigravity's internal process, authentication flow, and API behavior to enable zero-configuration, secure local integration.
  - **Features:**
    - Real-Time Monitoring — Displays lowest available model quota directly in VS Code status bar
    - Dynamic Port Discovery — Automatically detects the running Antigravity process and resolves server port
    - Session Token Authentication — Extracts and reuses active session tokens (no manual API key required)
    - Auto & Manual Refresh — Background polling (60s) + on-click refresh
    - Quota Intelligence — Categorizes model health (healthy, low, exhausted) with visual indicators
    - Secure Local Networking — Handles localhost SSL bypass for self-signed certificates
    - Detailed Tooltip UI — Shows full model breakdown (Gemini, Claude, GPT variants, etc.)
    - Robust Error Handling — Handles process detection failures, zero quotas, and connection states
  - **Engineering Highlights:**
    - Reverse engineered Antigravity local server communication flow
    - Implemented process scanning via PowerShell to dynamically locate active instances
    - Built lightweight status bar UI using VS Code Extension API
    - Designed efficient polling mechanism with minimal performance overhead
    - Structured modular TypeScript architecture for maintainability
  - **Technology:**
    - TypeScript, Node.js, VS Code Extension API, PowerShell Process Detection, Local HTTPS Handling, REST API Integration
  - **Role:**

<table>
<tr><td></td><td>

- Sole Developer — Architecture, Reverse Engineering, API Integration, UI/UX, Packaging, Publishing

- **Publishing:**
  - Published on Microsoft VS Code Marketplace
    - **https://marketplace.visualstudio.com/items?itemName=AldrinJoanPandianW.gravigauge**
  - Published on Open VSX Registry
    - **https://open-vsx.org/extension/AldrinJoanPandianW/gravigauge**
  - Complete extension packaging, semantic versioning, and release management

</td></tr>
<tr><td>End-to-end Developer</td><td>

- **Bollywood Bias Buster(June 30, 2025 – June 19, 2025)**
  - **Description**: Detect, quantify, and remediate gender bias in Bollywood media. Classifies script dialogues by stereotype, rewrites biased text, visualizes bias trends, and generates structured PDF/CSV reports. Extended to posters (LLaVA-1.5) and trailers (emotion–gender plots) with Wikipedia-based linguistic and network analytics.
  - **Features**:
    - Data Cleaning — Extract gender-tagged dialogues from scripts (PDF → CSV) using NLP-based character and gender identification
    - Classification — Detect gender stereotypes with local Mistral-7B (GGUF)
    - Visualization — Bias category distribution, gender-wise trends
    - Rewrite — Stereotype-free rewriting using LLM guidance
    - Report Generation — Downloadable PDF + CSV reports
    - Poster Analysis — Visual bias detection with LLaVA-1.5
    - Trailer Analysis — Emotion–gender timeline plots
    - Wikipedia Analytics — Adjective/verb usage, centrality, coreference, songs metadata
  - **Technology**: Streamlit, Hugging Face Transformers (Mistral-7B, LLaVA-1.5), PyMuPDF, Pandas, Matplotlib, Seaborn, NetworkX, NLP pipelines
  - **Team Project**: Individual
  - **Role:** End-to-end Developer — bias classification, multimodal integration, UI/UX, and report generation
  - **GitHub:** https://github.com/Aldrin-Joan/Bolly_Bias_Streamlit_By_Aldrin_Joan_Pandian.git

</td></tr>
<tr><td>Android App & Large File Processing using AI & ML</td><td>

- **Legal AI Android & Web App** (March 29, 2025 – Ongoing)
  - **Description**: Developed a cross-platform AI legal assistant for Android and web. Enabled users to post legal issues, get AI-generated insights, access legal documents, and receive real-time legal updates with misinformation checks. Included voice-based multilingual chatbot and clause analysis.
  - **Technology**: React.js, Flutter, Django, FastAPI, MongoDB, Pegasus Indian Legal, CUAD, SpaCy, NLTK, Google STT/TTS, TD-IDF + LDA, AWS, Firebase, LangGraph
  - **Team Project:** 4 Members
  - **Role**: Front End (ReactJS), Flutter App Developer and Large File Processing using AI & ML

</td></tr>
<tr><td>Flutter Developer</td><td>

- **RehabEase – AI-Powered Rehabilitation Platform** (Dec 20, 2024 – May 26, 2025)
  - **Description**: Designed a mobile and web app for guided rehab exercises. Integrated motion tracking, real-time feedback, and progress monitoring.
  - **Technology**: Python Flask, Dart, Flutter, Firebase and Computer Vision
  - **Team Project**: 3 Members
  - **Role**: Team Leader and Android and IOS App Developer

</td></tr>
<tr><td>Front End</td><td>

- **Deep Learning-Based Knee Osteoarthritis Diagnosis** (Jan 29, 2025 – April 14, 2025)
  - **Description**: Implemented CNN and ensemble models to classify OA severity from X-rays. Added Grad-CAM and a web interface for real-time diagnosis.
  - **Technology**: Python, CNN, ReactJS, Machine Learning
  - **Team Project**: 6 Members
  - **Role**: Front End Developer

</td></tr>
<tr><td>Model Training</td><td>

- **Heart Disease Prediction Using Machine Learning** (Sep 23, 2024 – Dec 18, 2024)
  - **Description**: Created a machine learning model to predict heart disease risk. Applied feature engineering, classification, and visualization techniques.
  - **Technology**: Python, Jupyter, Machine Learning
  - **Team Project**: 2 Members
  - **Role**: Team Leader and Model Training

</td></tr>
<tr><td>Model Training</td><td>

- **Hand Sign Language Translation Using CNN** (Oct 21, 2024 – Dec 18, 2024)
  - **Description**: Built a CNN-based model to convert hand signs to text and vice versa. Enhanced gesture recognition using computer vision for accessibility.

</td></tr>
</table>

	- **Technology**: Python and CNN Model - **Team Project**: 5 members - **Role**: Model Training
Front End	▪ **AI-Powered Mentor-Mentee Platform** (March 14, 2024 – Oct 23, 2024) - **Description**: Developed a web app with AI-based matching to connect mentors and mentees. Integrated real-time chat, video calls, and calendar booking. - **Technology**: NextJS, Django, Firebase, MySQL, Python - **Team Project**: 6 Members - **Role**: Team Leader and Front-End Developer

INTERNSHIP	
Docu3C (Seattle, Washington, USA) (July 21, 2025 – Till Date)	**AI Engineer** • Designing and optimizing effective prompts for large language models to improve accuracy and reliability. • Building **multi-agent systems** using **LangGraph, LangChain, and LangSmith** to enable intelligent, collaborative AI workflows. • Applying advanced **NLP techniques** such as **TF-IDF, LDA, and NMF** for text analysis, topic modeling, and semantic search. • Developing and deploying **AI applications on Azure**, focusing on scalability, performance, and seamless integration. • Working extensively with **GPT-5** to create context-aware applications involving **file processing, automation, and real-time data handling**. • Debugging and optimizing **agentic systems**, ensuring reliability, error handling, and efficient orchestration. • Currently developing a **Travel Agent AI system** that integrates planning, weather updates, transport options, and real-time tracking through coordinated multi-agent workflows using Langgraph. **Skills** • Prompt Engineering, Multi-Agent Systems, LangChain, LangGraph, LangSmith • Natural Language Processing (TF-IDF, LDA, NMF, Semantic Search) • Generative AI (GPT-5, LLM fine-tuning, context management) • Azure Deployment, Cloud-based AI applications • File Processing, Automation, Real-time Data Handling • Debugging and Optimization of Agentic Systems
amasQIS.ai (Omen) (April 28, 2025 – October 31st, 2025)	• Worked as part of the Front-End team for the Personal Protective Equipment Compliance Project, contributing to webpage design and UI development. • Served as a Backend Developer in the Predictive Maintenance Project, integrating the YOLO model to meet custom requirements for defect detection. • Acting as Lead Flutter Developer for two projects, • LLM Workspace Adviser and VLM Workspace Adviser • Handling end-to-end mobile application development.
ApexPlanet Software Pvt. Ltd. (Oct 20, 2024 – Dec 4, 2024)	• Completed structured tasks involving webpage creation, responsive design, JavaScript interactivity, API integration, and localStorage. • Developed a dynamic to-do list, image gallery, and an E-commerce website as the final project. • Enhanced frontend development skills and implemented real-time data handling and performance optimization techniques.

CO-CURRICULARS	
Coding	▪ http://www.skillrack.com/profile/452987/63a1b8e881639ac762c3e1187e2cdf1f0f66308a ▪ 87600 Rank

EXTRA-CURRICULARS AND ACHIEVEMENTS	
Achievements	▪ Research Paper Presented, 3rd International Conference ICRP-2024 – ICRP (2024)

	▪ Attended, CSIR Sponsored National Seminar on Vehicular Cloud Model – CSIR (2024)
Responsibilities	▪ UGC NEP SAARTHI – Student Ambassador (Jan 02, 2025 – Till Date)
Extracurricular	▪ Completed 7 Grades in Keyboard – London College of Music (LCM) & Trinity College of Music

ADDITIONAL INFORMATION	
Research	▪ Research Paper Published, Al-Driven Predictive Maintenance for Smart Grid Components Architecture Applications, Challenges and Opportunities, Springer, 2024. ▪ Book Chapter Published, The Role of Artificial Intelligence in 6G Networks, Architecture, Protocol, Transmission and Applications, IGI Global, 2025. ▪ *Optimizing Video Compression for Real-Time and Bandwidth-Constrained Applications in IoT Enabled Cyber-Physical Systems* ▪ *Machine Learning Approaches to Support Corporate Environmental Governance through Accurate Emission Forecasting, Carbon Offset Allocation, and Green Fund Optimization* ▪ A Next-Generation Computing Approach for Early and Accurate Lung Cancer Detection Using an Integrated Hybrid Deep Learning Framework
Hobbies	▪ Reading Japanese Mangas ▪ Playing Games ▪ Playing Basketball
Languages	▪ English, Tamil, Hindi, Japanese